

Lab 4 Part I - Profiling MPI Code

Large Scale Hybrid Computing

University of Minho

The Lab 4 focus on the profiling of MPI code. Use a Deucalion node with ARM CPUs (by specifying the partition `normal-arm` when submitting a job). Compile the code either in the frontend or on the compute nodes (the latter is advised).

This lab tutorial includes five exercises to be solved during the lab class. Copy the source files, as usual, from `/projects/F202500010HPCVLABUMINHO/LSHC/lab4` to your home directory in Deucalion:

```
cp -r /projects/F202500010HPCVLABUMINHO/LSHC/lab4 .
```

To load the compiler in the environment use the following commands (do not forget to also use them in your submission script):

GNU Compiler: `module load GCC/13.3.0.`

MPI Libraries: `module load OpenMPI/5.0.3-GCC-13.3.0.`

Nvidia Profiler: `module load nvidia-compilers/25.1-prof.`

You can use other GNU/OpenMPI combinations at your own risk, but ensure that OpenMPI is compiled with the correct version of the compilation toolchain. Use `nsys` command line tools to profile the MPI code:

```
nsys profile --trace=mpi -o profile_report mpirun -np 2 ./diffusion
```

1. A simple profile with Nsight

Goals: to understand the profiling experience with nsight.

Lab 1.1 The profiling workflow with Nvidia Nsight on supercomputing environments assumes that tracing happens in a CLI, with the reports stored for later analysis. Install Nsight on your system so that you can visualise the profiling results gathered in Deucalion. The binary is available at:

<https://developer.nvidia.com/nsight-systems/get-started>

Lab 1.2 Consider the provided code. It implements a 1D version of the heat diffusion code of lab1, with no real scientific value, which will be used to assess the capabilities of the profiler. Identify the MPI primitives used on this implementation.

Perform a basic profiling targeting the MPI component of the code using Nsight. It is crucial that you enable debugging symbols during the compilation process:

```
mpicc -O3 -g -lm heat_diffusion.c -o diffusion
```

Profile the code on 4 nodes using 4 processes per node. Create the appropriate job script (do not forget to set `ntasks`!), and profile the code:

```
nsys profile --trace=mapi -o profile_report_%q{SLURM_PROCID} mpirun -np 16 ./diffusion
```

Copy the profile trace to your system and open it with Nsight. Located on the profiling timeline the MPI primitives previously identified.

Lab 1.3 Explore the NSight visualiser. It contains two panels, a timeline and an event view for the current analysis. Try to answer the following questions:

- Identify the execution of the main loop in the processes.
- Identify how many threads are being used and their purpose.
- Identify the # bytes per message sent by `SSend` and `ISend` primitives.